Tutorial – 05

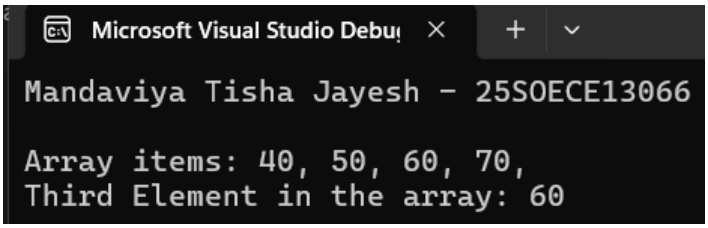
1. **Write a C# Sharp program that stores elements in an array and prints them.**

Code:

```
using System;

namespace Practicals
{
  class T5_01
  {
    public static void T5_01Main()
    {
      int[] arr = { 40, 50, 60, 70 };
      Console.Write("Array items: ");
      for (int i = 0; i < arr.Length; i++)
      {
        Console.Write(arr[i] + ", ");
      }
      //individual element print
      Console.WriteLine("\nThird Element in the array: " + arr[2]);
    }
  }
}
```

Output:



2. **Write a program of sorting an array. Declare a single dimensional array and accept 5 integer values from the user. Then sort the input in ascending order and display output.**

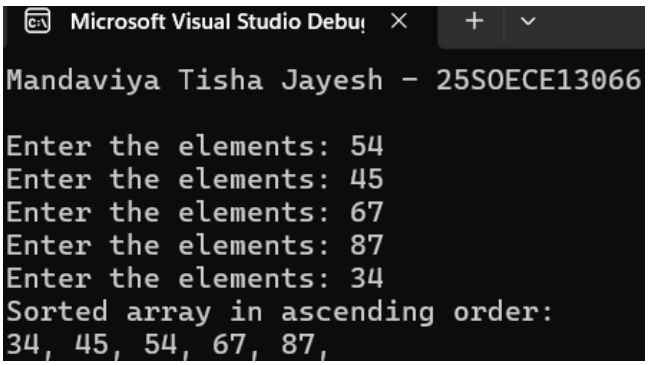
Code:

```
using System;

namespace Practicals
{
```

```
class T5_02
{
    public static void T5_02Main()
    {
        int[] arr = new int[5];
        for (int i = 0; i < arr.Length; i++)
        {
            Console.Write("Enter the elements: ");
            arr[i] = Convert.ToInt32(Console.ReadLine());
        }
        Array.Sort(arr);
        for (int i = 0; i < arr.Length; i++)
        {
            Console.WriteLine(arr[i]);
        }
    }
}
}
```

Output:



3. **Write a C# Sharp program to read n values in an array and display them in reverse order.**

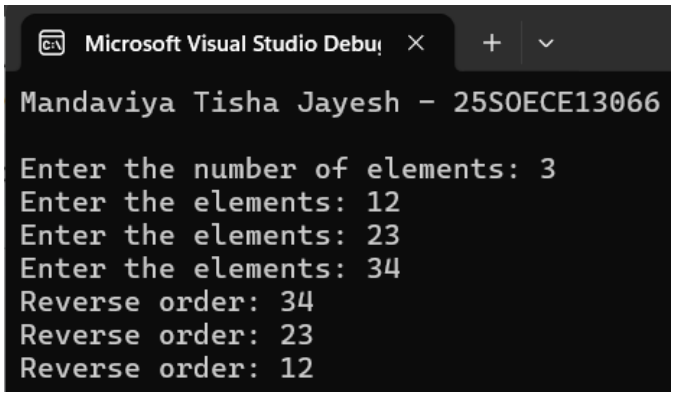
Code:

```
using System;

namespace Practicals
{
    public class T5_03
    {
        public static void T5_03Main()
        {
            Console.Write("Enter the number of elements: ");
```

```
            int n = Convert.ToInt32(Console.ReadLine());
            int[] arr = new int[n];
            for (int i = 0; i < n; i++)
            {
                Console.Write("Enter the elements: ");
                arr[i] = Convert.ToInt32(Console.ReadLine());
            }
            for (int i = n-1; i >= 0; i--)
            {
               Console.WriteLine("Reverse order: " + arr[i]);
            }
        }
    }
}
```

Output:



4. **Write a C# Sharp program to copy the elements of one array into another array.**

Code:

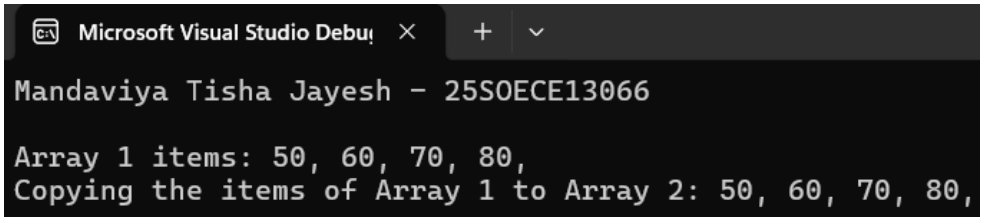
```
using System;

namespace Practicals
{
   class T5_04
   {
      public static void T5_04Main()
      {
         int[] arr1 = { 50, 60, 70, 80 };
         int[] arr2 = { 10, 20, 30, 40 };

         Array.Copy(arr1, arr2, arr1.Length);
         Console.Write("Array 1 items: ");
```

```
            for (int i = 0; i < arr1.Length; i++)
            {
                Console.Write(arr1[i] + ", ");
            }
            Console.Write("\nCopying the items of Array 1 to Array 2: ");
            for (int i = 0; i < arr2.Length; i++)
            {
                Console.Write(arr2[i] + ", ");
            }
            Console.ReadLine();
        }
    }
}
```

Output:



5. **Write a C# Sharp program to count duplicate elements in an array.**

Code:

```
using System;

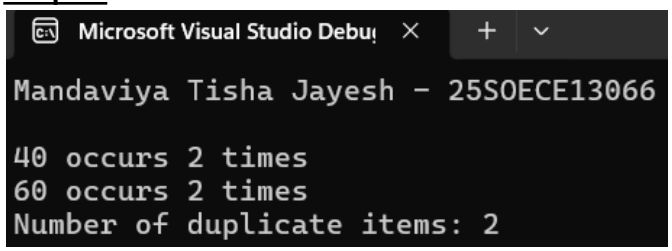
namespace Practicals
{
    class T5_05
    {
        public static void T5_05Main()
        {
            int[] arr = { 40, 50, 60, 70, 40, 60, 80 };
            int duplicateCount = 0;
            for (int i = 0; i < arr.Length; i++)
            {
                bool alreadyChecked = false;

                //check whether that element appeared before
                for (int j = 0; j < i; j++)
                {
```

```
                if (arr[i] == arr[j])
                {
                    alreadyChecked = true;
                    break;
                }
            }
            if (alreadyChecked)
                continue;

            int count = 0;
            for (int j = 0; j < arr.Length; j++)
            {
                if (arr[i] == arr[j])
                {
                    count++;
                }
            }
            if (count > 1)
            {
                Console.WriteLine(arr[i] + " occurs " + count + " times ");
                duplicateCount++;
            }
        }
        Console.WriteLine("Number of duplicate items: " + duplicateCount);
    }
  }
}
```

Output:



6. **Write a C# Sharp program to find the maximum and minimum elements in an array.**

Code:

```
using System;

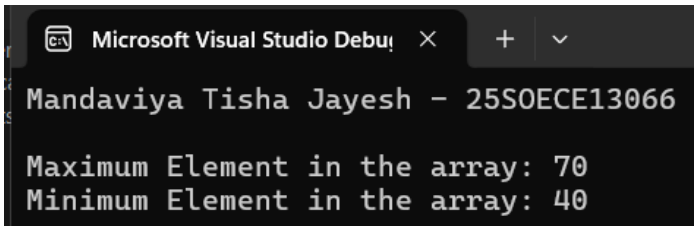
namespace Practicals
```

```
{
    class T5_06
    {
        public static void T5_06Main()
        {
            int[] arr = { 40, 50, 60, 70 };
            int max = arr[0];
            int min = arr[0];

            for (int i = 1; i < arr.Length; i++)
            {
                if (arr[i] > max)
                    max = arr[i];

                if (arr[i] < min)
                    min = arr[i];
            }
            Console.WriteLine("Maximum Element in the array: " + max);
            Console.WriteLine("Minimum Element in the array: " + min);
        }
    }
}
```

Output:



7. **Write a program in C# Sharp to separate odd and even integers into separate arrays.**

Code:

```
using System;

namespace Practicals
{
    class T5_07
    {
        public static void T5_07Main()
        {
            int[] arr = { 10, 15, 20, 25, 30, 35, 40 };
```

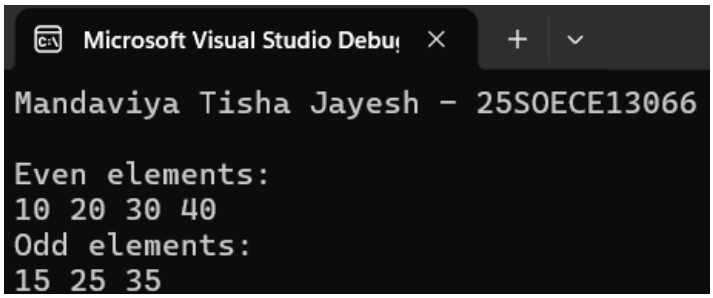
```
            int[] even = new int[arr.Length];
            int[] odd = new int[arr.Length];
            int e = 0, o = 0;

            for (int i = 0; i < arr.Length; i++)
            {
                if (arr[i] % 2 == 0)
                {
                    even[e] = arr[i];
                    e++;
                }
                else
                {
                    odd[o] = arr[i];
                    o++;
                }
            }
            Console.WriteLine("Even elements: ");
            for (int i = 0; i < e; i++)
            {
                Console.Write(even[i] + " ");
            }

            Console.WriteLine("\nOdd elements: ");
            for (int i = 0; i < o; i++)
            {
                Console.Write(odd[i] + " ");
            }

        }
    }
}
```

Output:



8. Write a C# Sharp program to sort array elements in descending order.

Code:

```
using System;

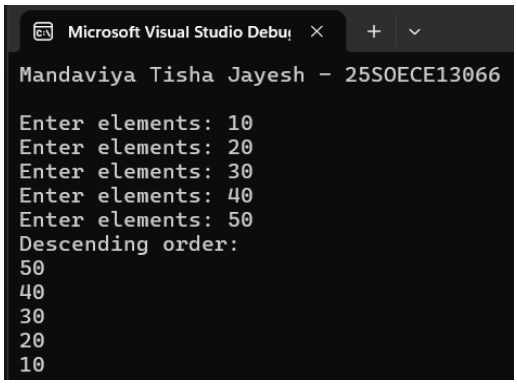
namespace Practicals
{
  class T5_08
  {
    public static void T5_08Main()
    {
      int[] arr = new int[5];

      for (int i = 0; i < 5; i++)
      {
        Console.Write("Enter elements: ");
        arr[i] = Convert.ToInt32(Console.ReadLine());
      }

      //Array.Sort(arr);
      //Array.Reverse(arr);

      Console.WriteLine("Descending order: ");
      for (int i = 4; i >= 0; i--)
      {
        Console.WriteLine(arr[i]);
      }
    }
  }
}
```

Output:



9. Write a C# Sharp program to delete an element at the desired position from an array.

Code:

```
using System;

namespace Practicals
{
   class T5_09
   {
      public static void T5_09Main()
      {
         int[] arr = { 10, 20, 30, 40, 50 };

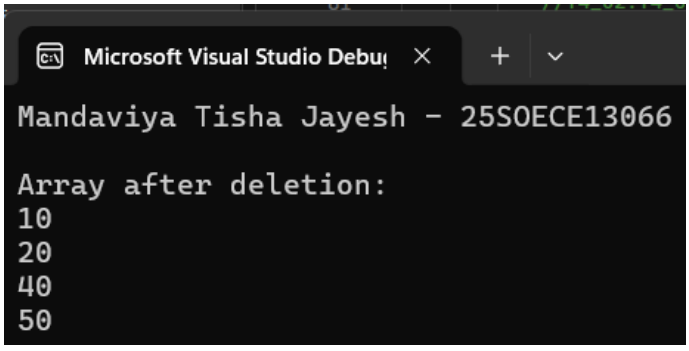
         int pos = 3;

         for (int i = pos - 1; i < arr.Length - 1; i++)
         {
            arr[i] = arr[i+1];
         }

         Console.WriteLine("Array after deletion: ");
         for (int i = 0; i < arr.Length - 1; i++)
         {
            Console.WriteLine(arr[i]+ " ");
         }

      }
   }
}
```

Output:



10. Write a C# Sharp program for adding two matrices of the same size.

Code:

```
using System;

namespace Practicals
{
   class T5_10
   {
      public static void T5_10Main()
      {
         int[,] matrix1 = new int[2, 2];
         int[,] matrix2 = new int[2, 2];
         int[,] result = new int[2, 2];

         Console.WriteLine("Enter elements of first matrix: ");

         for (int i = 0; i < 2; i++)
         {
            for (int j = 0; j < 2; j++)
            {
               matrix1[i, j] = Convert.ToInt32(Console.ReadLine());
            }
         }

         Console.WriteLine("Enter elements of second matrix: ");

         for (int i = 0; i < 2; i++)
         {
            for (int j = 0; j < 2; j++)
            {
               matrix2[i, j] = Convert.ToInt32(Console.ReadLine());
            }
         }

         for (int i = 0; i < 2; i++)
         {
            for (int j = 0; j < 2; j++)
            {
               result[i, j] = matrix1[i, j] + matrix2[i, j];
            }
```

```
            }

            Console.WriteLine("Resultant matrix after addition: ");

            for (int i = 0; i < 2; i++)
            {
               for (int j = 0; j < 2; j++)
               {
                  Console.Write(result[i, j] + "\t");
               }
               Console.WriteLine();
            }
         }
      }
   }
```

Output:

